WSPRadar.org reconstruction & comparison

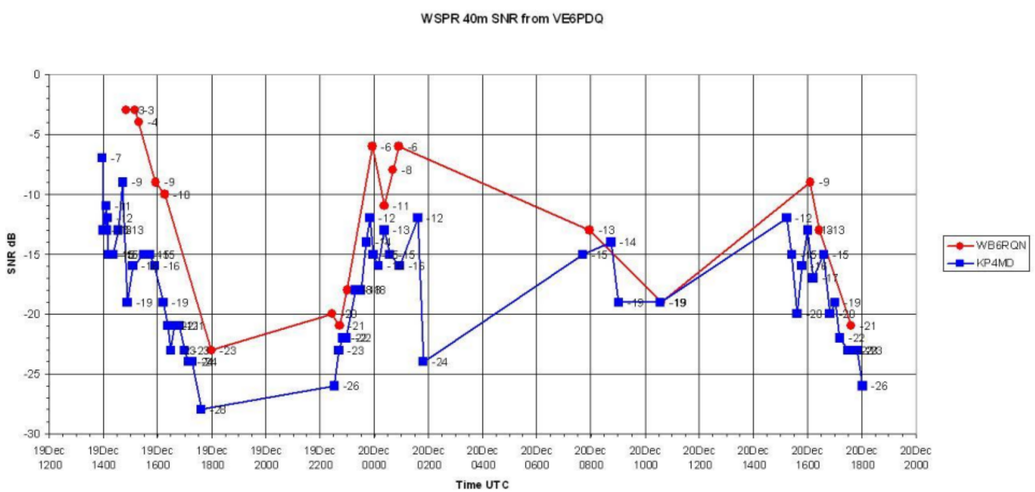
Reconciled TX direction: KP4MD / WB6RQN transmitting to VE6PDQ

Referenced publication: Carol F. Milazzo / KP4MD (2011). *Using the Weak Signal Propagation Reporter Network to Compare Antenna Performance.*

Source figure: **Figure 7** · Amateur-radio technical article and club presentation · 2011

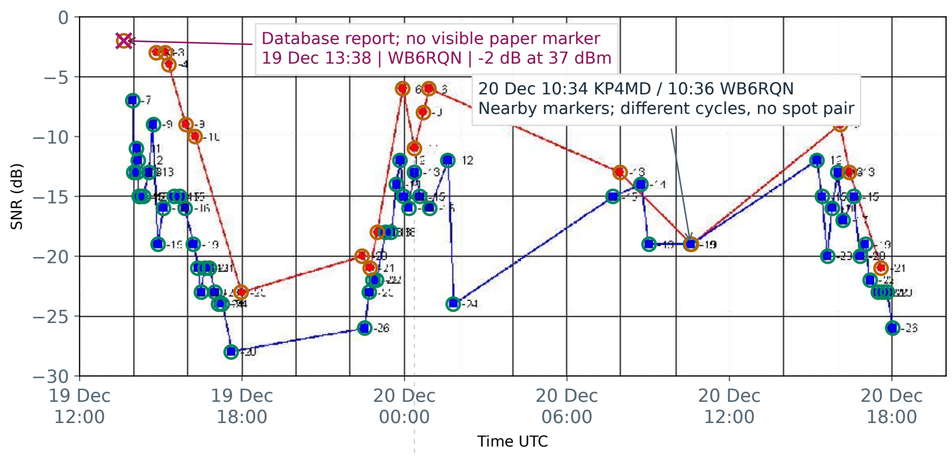
Demo: Historical TX replay | 40 m | 19 Dec 2010 12:00 to 20 Dec 2010 20:00 UTC | KP4MD / WB6RQN to VE6PDQ (DO34IR) | Reference correction 0 dB

Panel A - Original image from publication



Archive fields identify KP4MD / WB6RQN → VE6PDQ, opposite to the original direction label. The published image is retained unchanged.

Panel B — Archived reports on the published figure



76 archived reports; one spot pair

KP4MD is Target; WB6RQN is Reference.

Panel B overlays all 76 archived reports on the retained publication plot. The reconstruction identifies 47 published-point matches with agreement at the printed whole-dB SNR resolution. Archive timestamps are unchanged. Original lines supply no intermediate measurements.

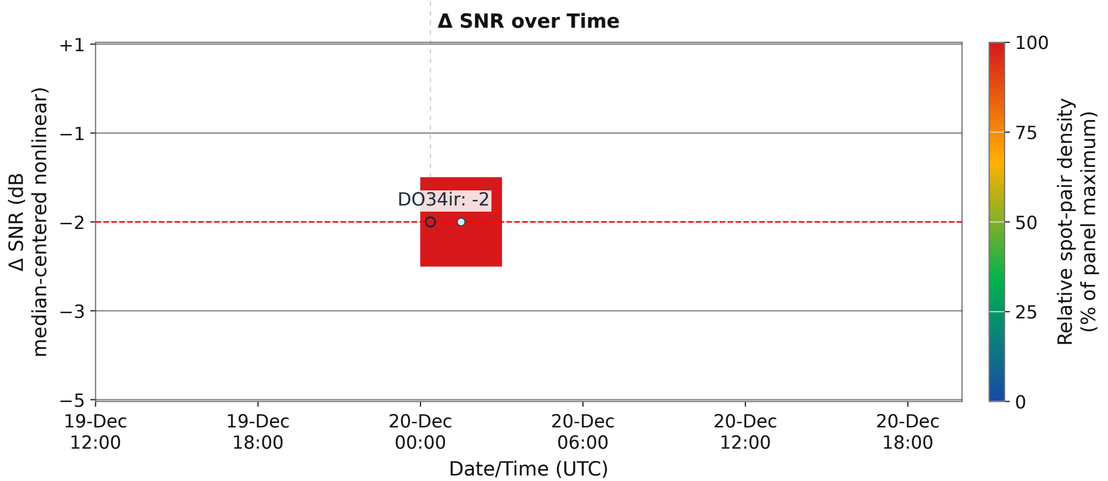
The inferred common 37 dBm scale leaves KP4MD's 5 W SNR unchanged and adds 4 dB to WB6RQN's 2 W reports. No additional fitted SNR offset is applied.

A spot pair requires the same remote callsign-and-locator entry to report both transmitters in the same two-minute WSPR cycle. Three-hour bins group the display only.

Only 20 December, 00:22 UTC qualifies: −13 − (−15 + 4) = −2 dB, favoring WB6RQN. One pair verifies this calculation, but establishes neither a typical advantage nor a time trend. Its observation and bin-median markers represent the same pair.

Panel C - WSPRadar paired ΔSNR

1 spot pair | KP4MD - WB6RQN | exact full locator identity



Circles: exact pairs; small markers: three-hour bin medians. Colour shows relative density, not confidence. The horizontal dashed line is the paired median. Read the nonlinear axis by its dB labels.

WSPRadar settings: peer distance < 5 000 km; minimum spot pairs per station: 1; all directions; solar state: all; special-callsign and moving-station exclusions: off.

WSPRadar.org by Dr. Markus Brosch (DL1MKS)

WSPRadar.org reconstruction & comparison

Observed TX coverage over time: counts and ΔDecode

Referenced publication: Carol F. Milazzo / KP4MD (2011). *Using the Weak Signal Propagation Reporter Network to Compare Antenna Performance.*

Source figure: **Figure 7** · Amateur-radio technical article and club presentation · 2011

Demo: Historical TX replay | 40 m | 19 Dec 2010 12:00 to 20 Dec 2010 20:00 UTC | KP4MD / WB6RQN to VE6PDQ (DO34IR) | Reference correction 0 dB

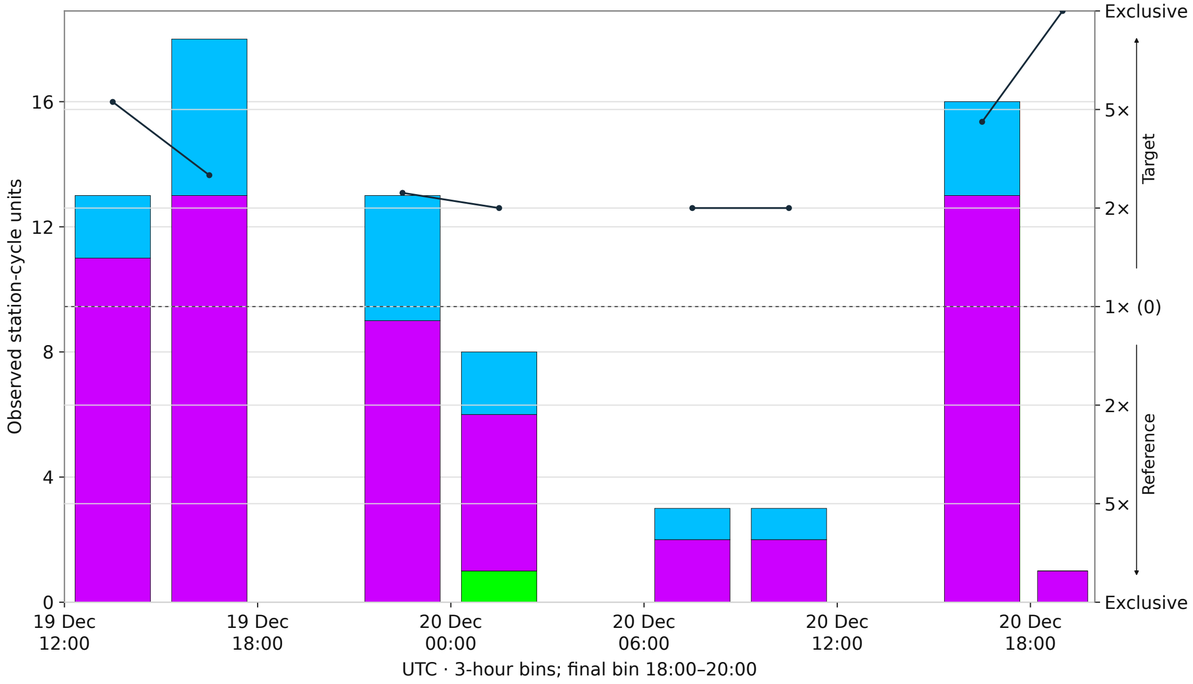
Overview · same-cycle, full-locator outcomes

Target: KP4MD; Reference: WB6RQN.

Selected identity	Only T	Spot pairs	Only R	Station-cycles	ΔDecode (×)
VE6PDQ / DO34IR	56	1	18	75	3.00×

Station-cycle: one observed remote callsign/locator/cycle event; timestamps may repeat.
Bars and outcome columns count events; fold factors compare report totals.

VE6PDQ / DO34IR · observed coverage



VE6PDQ / DO34IR · exact bin counts

UTC bin	Only T	Spot pairs	Only R	ΔDecode (×)
19 Dec 12:00–15:00	11	0	2	5.50×
19 Dec 15:00–18:00	13	0	5	2.60×
19 Dec 18:00–21:00	0	0	0	
19 Dec 21:00–00:00	9	0	4	2.25×
20 Dec 00:00–03:00	5	1	2	2.00×
20 Dec 03:00–06:00	0	0	0	
20 Dec 06:00–09:00	2	0	1	2.00×
20 Dec 09:00–12:00	2	0	1	2.00×
20 Dec 12:00–15:00	0	0	0	
20 Dec 15:00–18:00	13	0	3	4.33×
20 Dec 18:00–20:00	1	0	0	∞×

These 76 reports yield 57 for KP4MD and 19 for WB6RQN: 3×. They represent 75 observed station-cycles; the single spot pair contributes two reports. Counts retain the actual 5 W/2 W powers and operating schedules. SNR normalization does not change them; the 3× count ratio and −2 dB pair describe different evidence. Table factor = (Only T + Spot pairs)/(Only R + Spot pairs). 1×: equal; 0.5×: Reference 2×; ∞×: Target only; 0×: Reference only. Blank: no observations; transmission and decoding status unknown. Normalization cannot recover absent reports or equalize operating opportunities.